Photonic QDS Security Simulator

Detailed Technical and Cryptographic Engineering Specification

Project Title: Simulation-Based Quantum-Inspired Cyber Threat Detection for Long-Distance Photonic Quantum Digital Signatures
This document explains how the discrete-variable polarization-encoded photonic model, quantum teleportation protocol, classical security controls, long-distance optical channel modeling, and deterministic statistical threat detector function together as an integrated cryptographic system.
Primary Quantum Simulator: PennyLane v0.40+ | High-Performance Backend: PennyLane-Lightning | Optical Channel: NumPy + SciPy | Testing: Pytest (43/43 Tests Green) | Demo: Interactive Web Dashboard & CLI.

1. Project Objective & Core Architecture

The system is designed to detect both protocol-level cyber attacks and physical quantum-channel threats in a simulated long-distance photonic Quantum Digital Signature (QDS) system. Alice is the signer, Bob is the verifier/client, and Eve is the active adversary, communicating over an authenticated classical channel and a quantum optical fiber link (10 km to 200 km).

The key contribution is a deterministic, non-AI detection pipeline. Instead of training black-box neural networks or machine learning models (which are non-verifiable, subject to hallucination, and vulnerable to evasion), the simulator establishes legitimate measurement physics baselines and detects statistically significant anomalies using Total Variation Distance (D_TV), Pearson's Chi-Square tests (chi^2), and distance-aware adaptive thresholds.

2. How Classical and Quantum Together Form the QDS Protocol

A Quantum Digital Signature is fundamentally a hybrid classical-quantum cryptographic system. Neither classical cryptography alone nor quantum transmission alone can achieve information-theoretic digital signature guarantees:

- Why Classical Cryptography Alone Fails: Classical digital signature schemes (such as RSA, ECDSA, and Ed25519) rely on unproven computational complexity assumptions (e.g., Integer Factorization, Discrete Logarithms). These can be broken in polynomial time by Shor's algorithm on a quantum computer. Furthermore, classical bits can be perfectly copied, stored, and decrypted later ('Harvest Now, Decrypt Later').
- Why Quantum State Transmission Alone is Insufficient: The Quantum No-Cloning Theorem prevents an eavesdropper from copying unknown quantum states. However, raw photons cannot carry variable-length classical messages by themselves, cannot establish non-repudiation without cryptographic hash binding, and cannot survive long-distance fiber attenuation without classical feedforward correction.
- The Integrated Solution (Classical + Quantum Fusion): Classical cryptography provides message hashing (SHA-256), session tracking, replay protection (nonces), and sender identity authentication (HMAC-SHA256). Quantum mechanics provides the unforgeable signature payload via Pauli eigenstates, Bell entanglement (|Phi+>), teleportation feedforward, and multi-basis projective measurements (X, Y, Z) that physically expose any eavesdropping attempt.

3. Actors, Security Principals & Channel Topologies

Actor	Role	Operational Function & Cryptographic Actions
Alice	Signer	Possesses private key Key_Alice. Hashes message M, derives Pauli eigenstate sequence {|psi_k>}, performs Bell-state measurements (BSM), and initiates quantum state teleportation.
Bob	Verifier / Client	Receives classical metadata & photons, verifies classical freshness/HMAC tags, applies Pauli unitary corrections U = Z^c1 * X^c2, measures in X/Y/Z bases, and executes statistical threat detection.
Eve	Adversary	Attempts signature forgery, replaying stale nonces, impersonating Alice's identity, unauthorized verification, or intercepting/manipulating photons on the fiber link.

Two Complementary Channels:
1. Authenticated Classical Channel: Carries message text M, session ID, unique nonce, timestamp, HMAC-SHA256 authentication tag, and 2-bit BSM feedforward bits (c1, c2).
2. Photonic Quantum Optical Channel: Carries single polarization-encoded photons over modeled fiber distances L in {10, 25, 50, 100, 150, 200} km subject to attenuation, dephasing, and detector noise.

4. Photonic Polarization-Encoded Qubit Space

Quantum information is encoded in discrete-variable polarization modes of single photons in a 2-dimensional complex Hilbert space H_2. The six canonical Pauli eigenstates are structured across three mutually unbiased conjugate bases:

<table>
<tr><th>Basis</th><th>State</th><th>Polarization Mode</th><th>State Vector |psi></th><th>Density Matrix rho = |psi><psi|</th><th>Bloch (x, y, z)</th></tr>
<tr><td>Z (Comp.)</td><td>|0></td><td>Horizontal (H)</td><td>[1, 0]^T</td><td>[[1, 0], [0, 0]]</td><td>(0, 0, 1)</td></tr>
<tr><td>Z (Comp.)</td><td>|1></td><td>Vertical (V)</td><td>[0, 1]^T</td><td>[[0, 0], [0, 1]]</td><td>(0, 0, -1)</td></tr>
<tr><td>X (Diag.)</td><td>|+></td><td>Diagonal (D, +45 deg)</td><td>1/sqrt(2) [1, 1]^T</td><td>1/2 [[1, 1], [1, 1]]</td><td>(1, 0, 0)</td></tr>
<tr><td>X (Diag.)</td><td>|-></td><td>Anti-Diagonal (A, -45 deg)</td><td>1/sqrt(2) [1, -1]^T</td><td>1/2 [[1, -1], [-1, 1]]</td><td>(-1, 0, 0)</td></tr>
<tr><td>Y (Circ.)</td><td>|+y></td><td>Right-Circular (R)</td><td>1/sqrt(2) [1, i]^T</td><td>1/2 [[1, -i], [i, 1]]</td><td>(0, 1, 0)</td></tr>
<tr><td>Y (Circ.)</td><td>|-y></td><td>Left-Circular (L)</td><td>1/sqrt(2) [1, -i]^T</td><td>1/2 [[1, i], [-i, 1]]</td><td>(0, -1, 0)</td></tr>
</table>

The three conjugate bases probe complementary non-commuting observables. An eavesdropping attempt that minimizes perturbation in one basis produces maximum disturbance in the conjugate bases.

5. Quantum States, Bell Entanglement & Resource Budget

The primary entangled pair used for teleportation is the maximally entangled Bell state |Phi+> = (|00> + |11>)/sqrt(2) (photonic: (|HH> + |VV>)/sqrt(2)). The simulator also supports |Phi-> = (|00> - |11>)/sqrt(2), |Psi+> = (|01> + |10>)/sqrt(2), and |Psi-> = (|01> - |10>)/sqrt(2).

Exact Cryptographic Resource Budget per Signed Block:
• Signature Qubits (K): 16 qubits per message block (configurable: 8, 16, 32, 64).
• EPR Bell Pairs: 16 Bell pairs (|Phi+>) pre-distributed between Alice and Bob.
• Classical Feedforward: 32 classical bits (2 bits (c1, c2) per signature qubit).
• Measurement Shot Budget: 1,000 to 10,000 shots (N) per Pauli basis for statistical detection.

6. End-to-End System Flow & Classical Packet Structure

```
Message M (e.g., 'Transfer 1000 Quantum Credits to Bob')
    |
Alice / Signer ---> SHA-256 Digest + Session ID + Fresh Nonce
    |
Pauli Eigenstate Preparation {|psi_k>} ---> (0,0)->|0>, (0,1)->|1>, (1,0)->|+>, (1,1)->|+y>
    |
Bell Entanglement & Teleportation (BSM on (S, A))
    |------ Classical Channel Packet -----------------------------------------> Bob
    |       Header: SignerID='Alice', VerifierID='Bob', Version='1.0'
    |       Session: SessionID, Nonce (128-bit hex), Timestamp
    |       Payload: MessageText M, MessageHash H(M)
    |       Teleportation: BSM Bitstring 'c1_0 c2_0 ... c1_15 c2_15' (32 bits)
    |       Auth Tag: HMAC-SHA256(Key_Alice, SignerID:SessionID:Nonce:H(M):BSM_Bits)
    |
    `------ Photonic Fiber Link (10 - 200 km) ---> Attenuation/Noise ---> Eve ---> Bob
                                                                              |
                                           Bob Applies Pauli Correction U = Z^c1 * X^c2
                                                                              |
                                           Tier 1: Classical Cyber Auth & Freshness
                                                                              |
                                           Tier 2: QDS Quantum State Match
                                                                              |
                                           Tier 3: Multi-Basis X/Y/Z Measurements
                                                                              |
                                           Tier 4: Statistical Threat Engine (D_TV, chi^2)
                                                                              |
                                           Tri-State Decision: ACCEPT / SUSPICIOUS / REJECT
```

7. Step 1 — Message to Quantum Signature Mapping

Alice prepares message M. The digest is derived via SHA-256 and mapped to Pauli eigenstates: (0,0) → |0>, (0,1) → |1>, (1,0) → |+>, (1,1) → |+y>. Alice computes an HMAC-SHA256 authentication tag over (SignerID || SessionID || Nonce || H(M) || BSM_Bitstring) using her private key Key_Alice.

Implementation: `core/message.py` manages the message and hashing; `qds/signer.py` generates the signature; `quantum/pauli_states.py` defines the exact state vectors.

8. Step 2 — Quantum Teleportation Mechanics

Alice and Bob share an entangled Bell pair |Phi+>_AB. Alice combines target qubit S with her entangled half A into joint state |Psi_joint> = |psi>_S (x) |Phi+>_AB. Expanding into the Bell basis on (S, A):
|Psi_joint> = 1/2 [|Phi+>_SA (x) (alpha|0>+beta|1>)_B + |Phi->_SA (x) (alpha|0>-beta|1>)_B + |Psi+>_SA (x) (beta|0>+alpha|1>)_B + |Psi->_SA (x) (-beta|0>+alpha|1>)_B]

Alice performs Bell-State Measurement (BSM) yielding classical bits (c1, c2). Bob applies unitary Pauli correction U = Z^c1 * X^c2:
• Outcome 00: U = I -> Bob holds I|psi> = |psi> (Fidelity = 1.0)
• Outcome 01: U = X -> Bob applies X(X|psi>) = |psi> (Fidelity = 1.0)
• Outcome 10: U = Z -> Bob applies Z(Z|psi>) = |psi> (Fidelity = 1.0)
• Outcome 11: U = ZX -> Bob applies (ZX)(ZX|psi>) = |psi> (Fidelity = 1.0)
Ideal Reconstructed State Fidelity: F = 1.000000 +/- 0.000000 across all Pauli eigenstates.

9. Step 3 — Long-Distance Photonic Fiber Channel Model

The optical fiber link models physical photon propagation at 1550 nm across distances L in {10, 25, 50, 100, 150, 200} km:
• Fiber Attenuation: Transmittance T(L) = 10^(-alpha * L / 10) with nominal attenuation alpha = 0.20 dB/km (T(50 km) = 10.0%, T(100 km) = 1.0%, T(200 km) = 0.01%).
• Depolarizing Channel Noise: E(rho) = (1 - p)rho + (p/3)(X rho X + Y rho Y + Z rho Z) with distance accumulation p(L) = 1 - e^(-gamma * L).
• Single-Photon Detector Model: Detection efficiency eta = 0.85, dark count probability p_dark = 10^-5, and alignment jitter.

10. Why PennyLane and the Optical Model Are Decoupled

PennyLane acts as the primary quantum circuit engine, simulating state preparation, Bell-state entanglement, circuit execution, and Pauli corrections. The optical communication model (NumPy/SciPy) wraps the quantum states with physical channel effects (attenuation, thermal noise, dephasing, dark counts). This decoupling ensures physical channel modeling can be calibrated independently from circuit simulation.

11. Legitimate Baseline Calibration (P_{0,L})

Natural fiber degradation increases with distance. To prevent false alarms, the simulator pre-calibrates legitimate baselines P_{0,L} for each distance L. Under clean transmission, empirical baseline TVD is mu_{D_0} approx 0.035-0.038 across 10-200 km, ensuring legitimate fiber loss is never classified as an attack.

12. Multi-Basis Projective Measurements & Statistical Metrics

Bob samples received photons in X, Y, and Z bases over N shots, forming empirical distribution vector P_hat_N = [P(X+), P(X-), P(Y+), P(Y-), P(Z0), P(Z1)]^T.
• Total Variation Distance: D_TV(P_hat_N, P_{0,L}) = 0.5 * SUM |P_hat_N(k) - P_{0,L}(k)|.
• Pearson's Chi-Square Test: chi^2 = SUM (O_k - E_k)^2 / E_k.
• Distance-Aware Adaptive Threshold: tau(L, N, eta, alpha_sig) = mu_{D_0}(L) + z_{1-alpha} * (sigma_{D_0}(L) / sqrt(N)) + Delta_detector(eta).

13. Four-Tier Client Verification Architecture

Tier	Verification Subsystem	Evaluated Conditions	Failure Action
Tier 1	Classical Cyber Security	Signer HMAC Auth Tag, Nonce Store Freshness (Anti-Replay), Verifier ACL Authorization	Immediate REJECT (0 ms)
Tier 2	QDS Quantum State Match	Reconstructed state fidelity F_k = <psi_k|rho_k|psi_k>; Mismatch Rate R_mismatch ≤ 15%	REJECT (Forged Signature)
Tier 3	Statistical Threat Engine	Computes D_TV(P_hat_N, P_{0,L}), Pearson's chi^2, Z-scores vs adaptive threshold tau(L, N)	SUSPICIOUS or REJECT
Tier 4	Tri-State Arbitration	ACCEPT if D_TV ≤ tau_accept; SUSPICIOUS if tau_accept < D_TV ≤ tau_crit; REJECT if D_TV > tau_crit	Definitive Decision Record

14. Adversarial Threat Models & Defensive Guarantees

Attack Vector	Adversarial Operation	Detection Mechanism	Empirical Defense Rate
Signature Forgery	Eve generates random/guessed quantum states without Alice's private key	State fidelity mismatch rate exceeds 15% threshold	P_forge = 0.0000 (≤ 0.00209 bound)
Nonce Replay	Eve resends a previously valid classical/quantum signature bundle	Session & Nonce Store detects duplicate nonce hash	100.0% Rejection (30/30 blocked)
Signer Impersonation	Eve claims to be Alice using invalid or spoofed credentials	Entity Registry & HMAC-SHA256 public tag verification fails	100.0% Rejection (30/30 blocked)
Unauthorized Verifier	Rogue node attempts to verify Alice's signature without clearance	Verifier Authorization Access Control List (ACL) fails	100.0% Rejection (30/30 blocked)
Pauli Attacks (X, Y, Z)	Eve applies bit flip (X), phase flip (Z), or bit-phase flip (Y) on fiber	Statistical distance D_TV > adaptive threshold tau(L, N)	P_D = 100.0% for p_a ≥ 20%

15. Scientific Benchmark Experiments & Empirical Results

Exp #	Scientific Benchmark Suite	Key Empirical Benchmark Finding	Status
01	Teleportation Validation	Ideal fidelity F = 1.000000 +/- 0.000000 across all 6 Pauli states.	PASS (100%)
02	Photonic Channel Scaling	Adheres to alpha = 0.20 dB/km: T(50km)=10.0%, T(100km)=1.0%.	PASS
03	Legitimate Baseline P_{0,L}	Calibrated baseline TVD mu_{D_0} approx 0.035-0.038 across 10-200 km.	PASS
04	Pauli Attacks (X, Y, Z)	Detection probability P_D = 100.0% for attack strengths p_a ≥ 20%.	PASS (100%)
05	Signature Forgery Defense	Empirical P_forge = 0.0000 ≤ Theoretical Bound P_theo = 0.002090.	PASS (0% Forged)
06	Nonce Replay Defense	100.0% replay rejection rate (30/30 detected & dropped).	PASS (100%)
07	Signer Impersonation	100.0% spoofed signer rejection rate (30/30 detected).	PASS (100%)
08	Unauthorized Verifier	100.0% rogue verifier access rejection rate (30/30 detected).	PASS (100%)
09	Distance Sensitivity	Clean acceptance ≥ 96% up to 100 km; 100% attack detection.	PASS
10	Shot Count Scaling (N)	Statistical confidence converges cleanly from N=100 to N=10,000.	PASS
11	Threshold ROC Curve	Clear statistical separation between legitimate and attacked links.	PASS

16. Software Architecture, Testing & Execution Guide

Codebase Architecture: `quantum/` (circuits, teleportation, measurements), `photonic/` (polarization, fiber loss, noise, detectors), `qds/` (signer, verifier, signature logic), `security/` (authentication, freshness, integrity, ACL), `attacks/` (dispatch engine for Pauli, forgery, replay, impersonation), `detection/` (baselines, TVD, chi^2, adaptive threshold, decision engine), `visualization/` (dashboard, plotting suites).

```
# 1. Run full unit & integration test suite (43/43 tests green, 100% pass):
pytest tests/ -v

# 2. Launch interactive web demonstration dashboard (http://localhost:8000):
python main.py --demo

# 3. Execute all 11 scientific benchmark experiment suites:
python main.py --run-all-experiments

# 4. Run custom single simulation CLI:
python main.py --simulate --distance 50 --attack X --strength 0.20 --shots 1000
```

17. Important Scientific Boundaries & Conclusion

• Teleportation vs QDS: Teleportation is a physical state transfer mechanism; QDS incorporates signature generation, hash binding, classical authentication, and verification rules.
• 200 km Link: Modeled as a mathematical channel with parameterized loss and noise, not physical hardware.
• Deterministic Physics: Zero AI/ML black boxes guarantees mathematically provable, defensible security decisions.
• Verification Efficiency: Sub-30 ms latency and deterministic polynomial time complexity O(K * N) confirm practical computational feasibility.